



Tool to Isolate Air Intake Lines for Pre-Cleaner Washing Process

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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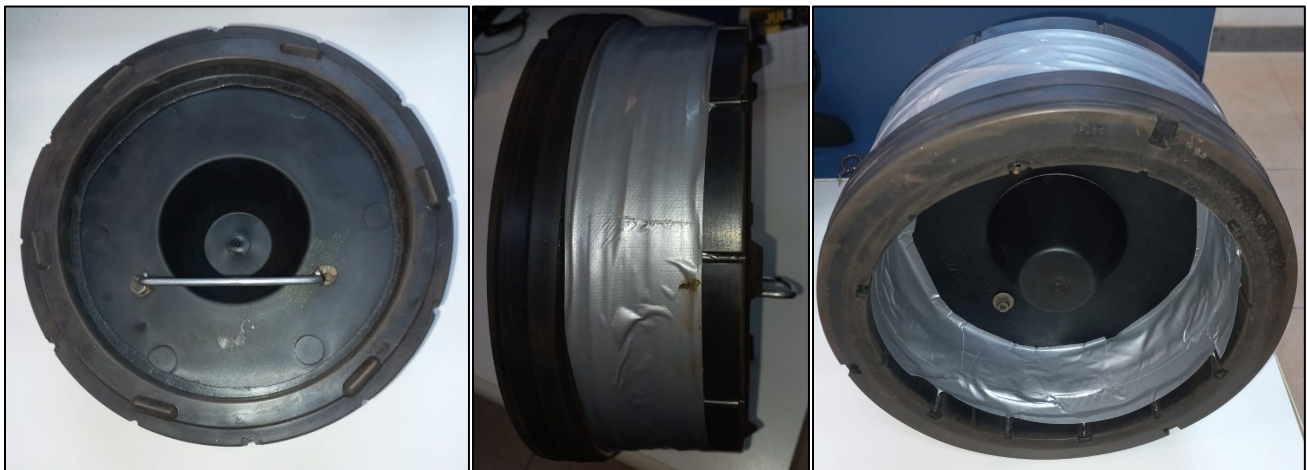
1.0 Introduction

The Operation and Maintenance Manual, SEBU8242, “789D and 789D XQ Off-Highway Trucks”, Engine Air Pre-Cleaner – Clean, recommends cleaning the pre-cleaner bowl with compressed air pressure and only when required. This compressed air cleaning process disperses dust in the environment and is not very efficient for cleaning the pre-cleaner: it contributes to the air filter clogging more quickly, increasing the changes of air filters that are changed by condition.

Due to these conditions and the risk of contamination by dust in the intake lines, a tool was developed with parts of the primary filter and parts of the secondary filter of the truck to isolate the intake lines, which prevents dust and water ingress during the preventive maintenance washing process. This condition leads to better cleaning of the pre-cleaner, thus increasing the estimated life of the primary and secondary air filters.

2.0 Best Practice Description

Tool made with parts of the secondary filter and the handle of the primary air filter of the 789D off-highway truck by cutting the parts that fit in the air intake line and creating the tool to isolate the line and seal the entrance of impurities and water from the air intake lines when cleaning the pre-cleaner.



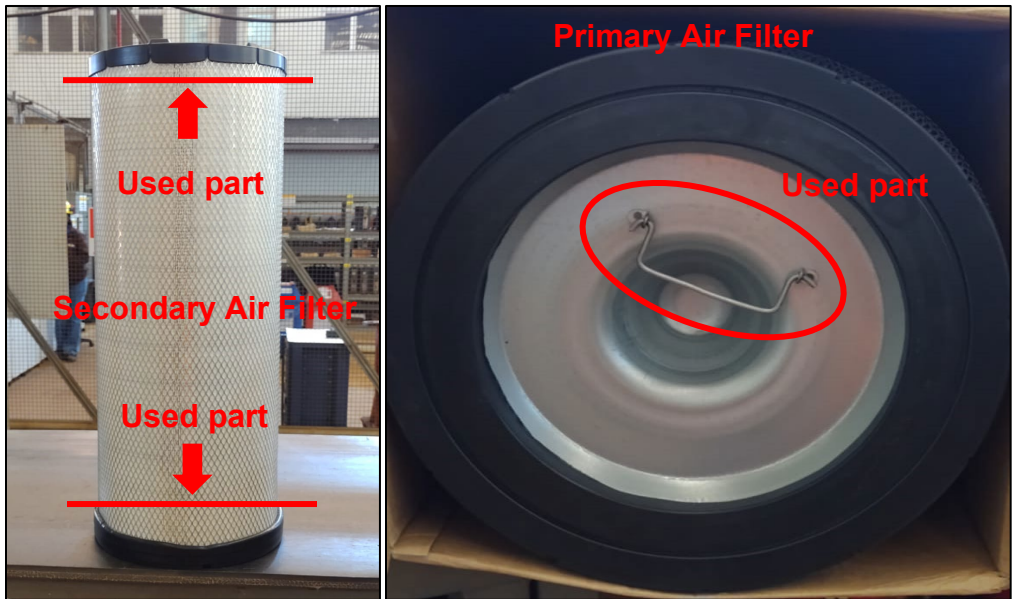
Picture 1: Tool to isolate air intake lines for pre-cleaner washing process.

The truck used as example in this Best Practice is a 789D SPD prefix, but the idea can be implemented in every machine that has similar pattern of air filters and pre-cleaner.

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3.0 Implementation Steps

- Select two pairs of primary and secondary air filters from the 789D truck that will be discarded.



Picture 2: Air Filter, Primary and Secondary.

- Clean, mark each side in a length of 50mm (indicated in blue) and cut the primary filter.



Picture 3: Cut the filter paper parts.

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- Using the cut parts of the secondary air filter, join the upper part to the lower part of the filter. Use silicone (wait until it dries) and waterproof tape (both sides, inside and outside of the tool).



Picture 4: Joining and sealing the parts.

- Remove the primary air filter handle. Using a drill with a 6mm drill bit, mark the distance from the handle, drill the plastic part, and install the handle with two 6mm screws in the tool made.



Picture 5: Handle installation.

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- Install tool in the air intake line;



Picture 6: Tool created installed in the air intake line.

- Wash the pre-cleaner with water at low pressure. Dry it with compressed air According to the procedure written in Operation and Maintenance Manual, SEBU8242, “789D and 789D XQ Off-Highway Trucks”, Engine Air Pre-Cleaner – Clean.

For 789Ds, fabricate two tools, one for the right side and one for the left side. For other machines in which this Best Practice can be implemented, fabricate a number of tools that requires the shortest time to complete the washing process.

4.0 Benefits

- Pre-Cleaner is cleaned in window of opportunity during the machine-washing process, eliminating downtime in this specific task (saving of 2 labor hours every 250 hours PM).
- Contamination control in air intake lines.
- Less contaminant accumulation on pre-cleaner inner walls with consequent.
 - I. Longer service life of air filters; and
 - II. Less probability of air filter restriction events.
- Less dispersion of dust in the working environment.

5.0 Resources Required

- Two used primary air filters
- Two used secondary air filters
- .Measuring tape
- Silicone
- Waterproof tape
- Drill, 6mm drill bit.

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6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

8.0 Acknowledgements

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